DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 24: The Lognormal Distribution of Prices

Shape & Moments · Quantiles · The Long-Horizon Paradox

September 28, 2026

Today's Agenda

1. The distribution of a future price

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2. Moments: mode, median, mean
3. Quantiles and confidence intervals
4. The long-horizon paradox
5. Why we model log-returns

Learning Objectives

- State the lognormal distribution of S_T and its parameters.

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- Explain the long-horizon paradox.

Lognormal Prices

Definition

$\ln S_T \sim \mathcal{N}(m, s^2)$ with

$$m = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T, \qquad s = \sigma\sqrt{T}$$

Normal on a Ruler That Stretches

What Exponentiating Does

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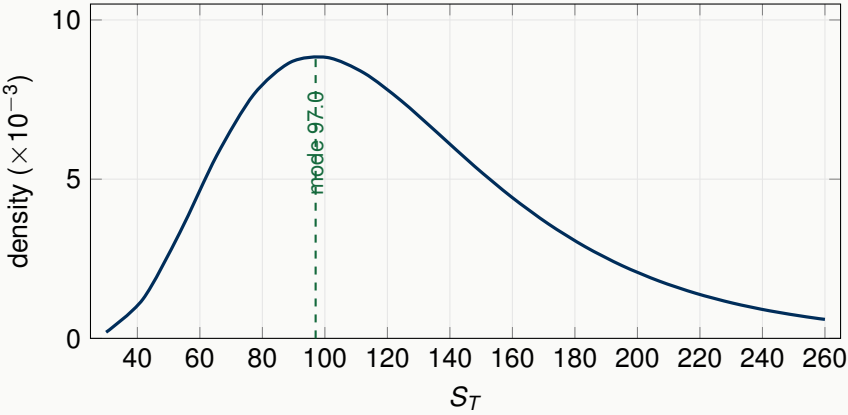
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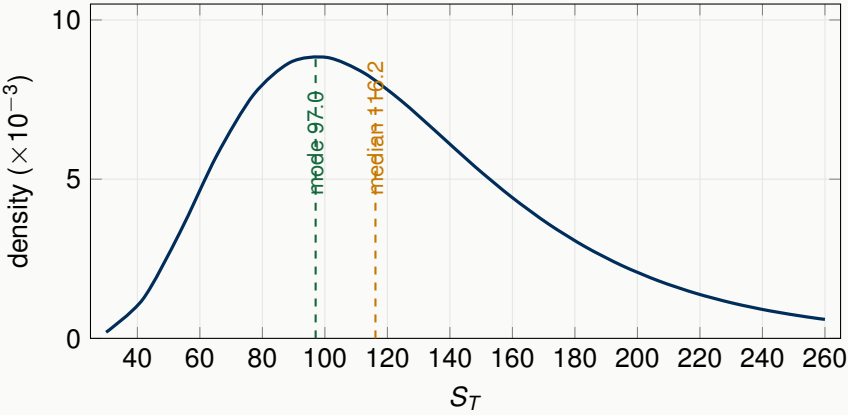
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Draw a bell curve on a rubber ruler, then stretch it exponentially. Distances near zero compress; distances far out pull apart. The result is symmetric *in log space*, but lopsided in price space: a hard floor at zero on the left, a long tail to the right. Not a quirk – a stock can only lose 100%, but can gain arbitrarily much.

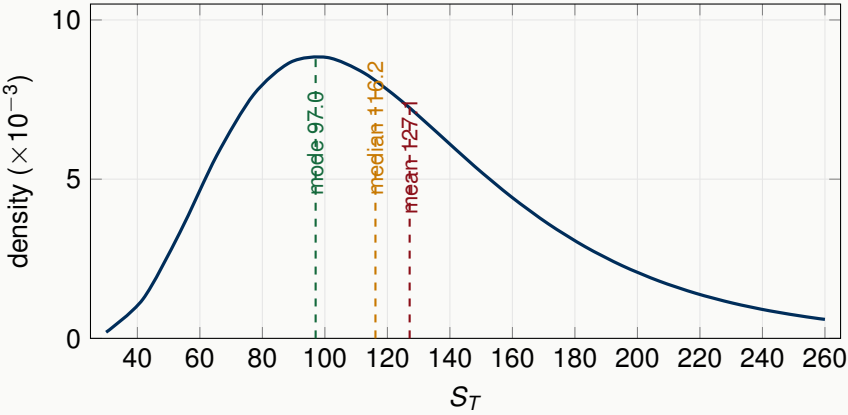
The Density


10
5
0
density ($\times 10^{-3}$)
mode 97.0
40
60
80
100
120
140
160
180
200
220
240
260
S_T

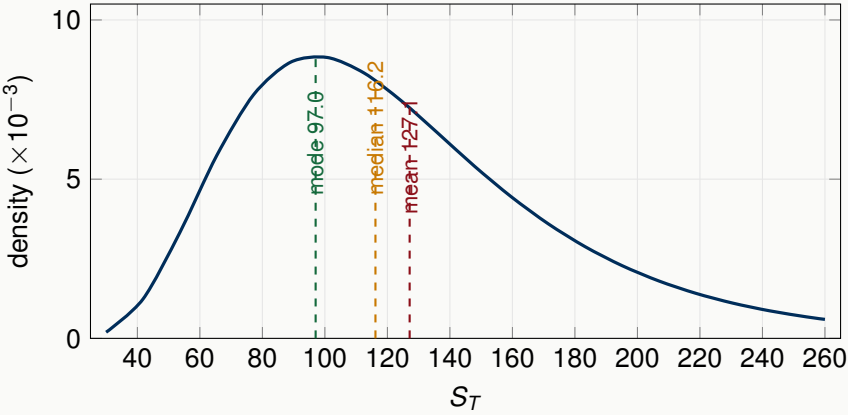
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S_T

The Density



S_0=100, μ=12%, σ=30%, T=2. Right-skewed.

The Four Key Quantities

Key Takeaway

$$\mathbb{E}[S_T] = S_0 e^{\mu T}, \qquad \text{median} = S_0 e^{(\mu - \sigma^2/2)T}$$

$$\text{mode} = S_0 e^{(\mu - 3\sigma^2/2)T}, \qquad \text{Var}(S_T) = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right)$$

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The ordering is always the same

$\mu - \frac{3\sigma^2}{2} < \mu - \frac{\sigma^2}{2} < \mu$, so

mode $<$ **median** $<$ **mean**

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always. Gaps widen with σ and T. **Quoting only the expected price overstates what a typical investor should expect.**

Worked Numbers

$S_0=100$, $\mu=12\%$, $\sigma=30\%$, $T=2$

	value	formula

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Median	116.18	$S_0 e^{(\mu-\sigma^2/2)T}$

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	value	formula
Mode	97.04	$S_0 e^{(\mu-3\sigma^2/2)T}$
Median	116.18	$S_0 e^{(\mu-\sigma^2/2)T}$
Mean	127.12	$S_0 e^{\mu T}$

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Std. dev.	56.45	—

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The mode is below the starting price of 100 – despite a $+12\%$ expected return.

Quantiles

Key Takeaway

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Percentile	5th	25th	50th	75th	95th
S_T	57.82	87.27	116.18	154.68	233.46

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90% interval $\approx [57.8,\ 233.5]$ – **wide, and strikingly asymmetric** about 100.

Symmetric in Logs, Asymmetric in Prices

Remark

In log space it *is* symmetric: $\ln(57.82/100) = -0.548$, $\ln(233.46/100) = +0.848$, both ± 0.698 around the log-median $+0.15$.

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General lesson: for multiplicative processes, work in logs; transform back only at the end.

The Long-Horizon Paradox

Mean Up, Median Down

If $\mu < \dfrac{\sigma^2}{2}$, then as $T \to \infty$:

$$\mathbb{E}[S_T] = S_0 e^{\mu T} \to \infty \qquad \text{but} \qquad \text{median} = S_0 e^{(\mu - \sigma^2/2)T} \to 0$$

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The average grows without bound, while *almost every path* drifts to zero.

A High-Volatility, Low-Return Asset

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After 50 years: the average more than quadruples; the typical outcome loses over 90%.

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The mean is propped up by a vanishingly small set of enormous successes.

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Why We Model Log-Returns

Key Takeaway

Under GBM, $r_T = \ln(S_T / S_0)$ is *exactly* normal. Practitioners use log-returns because:

- They're normally distributed (simple returns aren't – bounded below by -100%).

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- Variance scales linearly with time – the $\sqrt{T}$ rule from Week 3.

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- Quantiles: normal quantile in log space, then exponentiate.
- **Long-horizon paradox**: if $\mu < \sigma^2/2$, mean diverges while median collapses.
- **Log-returns**: normal, additive, variance linear in time.

Check Your Understanding

Discuss (2 minutes)

1. $S_0 = 200$, $\mu = 10\%$, $\sigma = 25\%$: find mean, median, mode of S_4.

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2. For $\mu = 6\%$: for which σ does the median *decline* over time?
3. Why is a log-utility investor immune to the long-horizon paradox?

Looking Ahead

Module 4 complete: a continuous-time price model, and the calculus to handle it.

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Black–Scholes–Merton – Week 9

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The number the binomial trees of Week 6 were converging to all along.